

Generality Claims and the Limits of Closed Optimization

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Abstract

Contemporary artificial-intelligence systems display broad and consequential capabilities. Those facts alone, however, do not establish that present methods are on a reliable path to artificial general intelligence (AGI). This Open Forum article argues for calibrated claims rather than for an impossibility thesis. Evidence from systematic-generalization and robustness research shows that high performance inside a defined evaluation regime need not establish reliable transfer to materially new compositions, contexts, or evidence conditions. Counterevidence matters: some training regimes can induce important forms of systematic generalization. The appropriate conclusion is therefore that generality remains an empirical target requiring direct evaluation, not a property licensed by scale, fluency, commercial value, or a collection of benchmark successes. The article develops a reciprocal-coupling hypothesis as one possible explanation of the remaining gap. It then offers a host-governed AI architecture as an inspectable research proposal: learned models generate possibilities, while external components preserve evidence, constrain assumptions, route work, and verify actions. This proposal is not presented as an implementation, a demonstration of cognition, or a solution to AGI. Finally, the article identifies a governance risk—*infrastructural entrapment*—through which institutions may come to depend on capable but bounded systems before their reliability boundaries are known. The practical recommendation is proportionate evaluation, plural research investment, and retained fallbacks for high-consequence deployment.

Keywords: artificial general intelligence; systematic generalization; robustness; cognitive architecture; AI governance; human–AI interaction

1. Introduction

Public discussion of artificial general intelligence often joins several different propositions. A model answers questions across many domains; a product is economically useful; benchmarks improve; therefore a reliable route to general intelligence has been established. Each observation may be true, but the inference between them is substantially stronger than the observations themselves. A system can be useful over a wide task range without reliably transferring to unfamiliar combinations, remaining calibrated outside familiar conditions, or reorganising itself in response to a changing environment.

This article argues for epistemic restraint about that inference. It does not argue that AGI is impossible, that neural networks cannot exhibit systematic behaviour, or that no future development of current methods could meet a demanding account of generality. Its more limited claim is that current evidence does not yet establish those conclusions. In particular, optimization against specified objectives and bounded training and evaluation distributions has not, by itself, demonstrated robust open-world generalization, systematic compositionality, or self-maintaining adaptation. According to the current evidence, it is therefore premature to treat scale, broad task coverage, or commercial deployment as sufficient evidence that these properties will emerge at the required level.

The distinction matters both scientifically and socially. Scientifically, a claim of general intelligence must be answerable to a specification of the generality, reliability, and autonomy being claimed. Morris et al. (2023) make this point by separating performance, breadth of generality, and autonomy in their proposed terminology. Socially, language that blurs demonstrated scope and projected generality may affect procurement, oversight, labour design, and public expectations before the relevant systems have been tested under the conditions in which they will be relied upon.

The argument is organised as follows. Section 2 states its evidential standard and limits. Section 3 considers what compositional-generalization and robustness results establish, and what they do not. Section 4 develops reciprocal coupling as a hypothesis rather than a settled theory of cognition. Section 5 specifies a host-governed design proposal and the claims it would have to earn empirically. Section 6 treats a recent biological theory as a nonvalidating functional comparison. Sections 7 and 8 draw out the governance implications and a research programme. The paper's contribution is thus a conceptual and methodological argument: claims of generality should be calibrated to demonstrated reliability, while alternative system organisations should be evaluated rather than ruled in or out rhetorically.

2. The Claim, Its Evidential Standard, and Its Limits

The term *generality* is used in many ways. At a minimum, it should not mean merely that a system has been prompted on many topics. A minimally demanding account includes reliable transfer to materially new combinations, sensitivity to task-relevant structure rather than only familiar surface regularities, and calibrated acknowledgement of uncertainty when the available evidence does not warrant a confident response. More demanding accounts may add autonomous learning of new skills, continuing adaptation, embodiment, goal formation, or self-maintenance. This article does not attempt to settle which full account is correct. It uses the narrower account to establish a point that holds across many stronger ones: finite, favourable demonstrations do not by themselves show dependable generality beyond their tested conditions.

The relevant distinction is between a capability observation and a generality inference. A capability observation has a defined scope: under stated data, tasks, interfaces, tools, and scoring rules, a system achieved a stated result. A generality inference extends that observation: it proposes that the system will transfer reliably to materially different combinations or environments. The former can be strong evidence. The latter requires direct support. The error is not using benchmarks; it is treating a benchmark as though it measured properties it was not designed to measure.

This standard excludes two symmetrical mistakes. The first is triumphalism: inferring an unmeasured open-world capacity from a set of striking demonstrations. The second is architectural fatalism: inferring from documented failures of one model, benchmark, or training recipe that neural systems as such cannot be systematic. Lake and Baroni (2023), for example, report a meta-learning regime in which a neural network acquired important forms of human-like systematic generalization. That finding is a material counterweight to sweeping claims about neural networks. It also does not show that ordinary next-token training at scale, or any particular deployed model, reliably

possesses the same property under changed environments. Both the positive and negative results call for evaluation that states exactly what was trained, tested, and transferred.

The article is also not an empirical report. It introduces no new benchmark result, and it does not claim causal confirmation of the proposed architecture in Section 5. Where it discusses empirical work, the claim is restricted to what the cited studies test. Where it discusses reciprocal coupling, host governance, or infrastructural entrapment, the relevant status is hypothesis, design proposal, or risk mechanism. Keeping those statuses separate is necessary to avoid turning an attractive architecture into purported evidence that it works.

3. What Present Evidence Does and Does Not Establish

3.1 Systematic generalization

The systematicity concern is longstanding. Fodor and Pylyshyn (1988) asked whether a cognitive system that can handle familiar structured expressions can handle related novel combinations for principled rather than merely associative reasons. The contemporary relevance of that question is not that it settles the nature of intelligence in advance. It is that a system can achieve high average performance where training and test distributions overlap while failing when the same components are recombined in a new structural arrangement.

SCAN was designed to make this distinction visible. Lake and Baroni (2018) found that standard sequence-to-sequence recurrent models learned familiar mappings but failed on demanding compositional splits. COGS extended the challenge to semantic interpretation. Kim and Linzen (2020) reported very high in-distribution performance for Transformer and LSTM models alongside considerably weaker performance on the compositional generalization split. These experiments are valuable precisely because they do not ask whether a model is impressive in general; they test a specified form of transfer.

The correct reading is modest. SCAN and COGS do not prove that every contemporary model fails at every compositional task. They do not measure all possible forms of intelligence, and particular benchmark designs can reward or penalise different strategies. Nor are early recurrent-model findings automatically transferable to every later architecture or training regime. What they establish is a methodological warning: strong performance under familiar distributions can coexist with a material gap in a targeted form of structural transfer. A claim that a system is generally reliable must therefore test the transfer at issue instead of assuming it from interpolation.

Subsequent work makes the same point from the other direction. Lake and Baroni (2023) show that training structure can substantially alter systematic-generalization outcomes. This is a reason to investigate the conditions under which the property arises, not a reason to declare it either solved or impossible. Goodale and Mascarenhas (2026), an unreviewed preprint, argue that the original systematicity challenge remains open. It should be treated as one current intervention in an unsettled debate, not as decisive proof. Taken together, the literature supports neither a categorical verdict nor a smooth extrapolation from general-purpose language performance. It supports the more prosaic conclusion that systematicity must be directly operationalised and measured.

3.2 Robustness, distribution shift, and evidential conditions

The issue is wider than linguistic composition. Geirhos et al. (2020) describe shortcut learning: systems may exploit predictive regularities that support high benchmark performance but fail when those regularities change. Gulrajani and Lopez-Paz (2021) similarly find no uniformly dominant method across their domain-generalization settings. Neither paper proves that all machine learning is brittle, or that deployment is futile. They document a recurring reason to keep performance claims conditional on the distribution, environment, and objective under which they were measured.

For language-model systems, the conditions that matter may include prompt framing, tool availability, retrieved sources, context length, task decomposition, human oversight, and the practical consequences of a mistake. A model-plus-tool system may outperform a model alone on a defined task. That is a meaningful design fact, but it is not evidence that the assembled system will be reliable wherever the label “general” is applied. Conversely, a failure under one prompt or benchmark does not establish a universal limitation. The relevant unit of assessment is the deployed socio-technical system, with its interfaces, inputs, safeguards, users, and fallback procedures specified.

This is closely aligned with the National Institute of Standards and Technology’s AI Risk Management Framework. NIST treats validity, reliability, robustness, and monitoring as context-sensitive properties requiring ongoing attention rather than traits inferred once and for all from a model label (Tabassi 2023). A context-sensitive framework is not an anti-innovation rule. It is a demand that confidence track the conditions under which a system’s output will matter.

3.3 What follows—and what does not

The empirical premises support three conclusions. First, breadth of task performance is not identical with tested structural transfer. Second, present results make it unreasonable to assume that scale or fluency alone supplies the missing transfer. Third, training and system organisation can change outcomes, so the open question is constructive as well as critical. They do not support a numerical estimate of AGI’s probability, a declaration that current approaches cannot improve, or a prediction that every deployment will fail.

This distinction is important because the phrase “current evidence suggests” is often used as an unnoticed substitute for “current evidence demonstrates.” Here it means only that the cited findings are consistent with a live uncertainty about generality and justify direct tests of it. The argument reaches no farther than that. A credible positive claim would need comparative evidence showing reliable transfer, calibration, recovery, and adaptation across relevant changed conditions—not merely a larger collection of familiar successes.

3.4 Evaluation is a measurement problem, not a rhetorical contest

The demand for direct testing can sound obvious while remaining underspecified. A useful evaluation should state at least four things. It should state the system boundary: whether the evaluated object is a model, a model with retrieval, a model with tools, a workflow with human reviewers, or a larger operational arrangement. It should state the source and timing of task information, including whether a system can depend on information that would be

unavailable in deployment. It should specify the kind of change being tested: lexical novelty, compositional novelty, a new domain, a changed interface, conflicting evidence, an altered incentive, or a different consequence of error. Finally, it should state what counts as a successful response: completion, truth, calibrated deferral, recoverability, traceability, or an appropriate combination of these.

Without those specifications, both promotional and critical examples can become misleading. A demonstration may look general because it spans visible domains while preserving the same interaction pattern, source conventions, and evaluator support. A failure may look fundamental because it occurs in an artificial task split that does not resemble any material use. The remedy is not to dismiss either kind of evidence. It is to connect the evaluation change to the deployment claim. If an organisation proposes to rely on a system for decisions involving incomplete records and conflicting sources, the relevant test is not only a clean benchmark with one correct answer. If it proposes to automate a reversible drafting task, failure under a high-stakes autonomous-action scenario may have limited relevance.

The same discipline applies to claims about emergent capacities. An unanticipated behaviour may be scientifically interesting, but surprise is not a reliability measure. A capacity becomes useful for a generality claim when it is reproduced, bounded, compared against alternatives, and shown to persist under relevant changes. Conversely, an absence of a capacity in today's tests does not prove that it cannot be developed. The article's objection is thus to premature closure in either direction: no capability should be credited merely because it is rhetorically impressive, and no research path should be ruled out merely because current evidence is incomplete.

4. The Reciprocal-Coupling Hypothesis

A contrast between top-down engineering and bottom-up emergence would be misleading. Biological development, evolution, learning, institutions, and engineered systems all involve constraints and control as well as local interaction. The more discriminating contrast is between *closed, unidirectional optimization* and *reciprocal coupling*.

In a closed optimization regime, a system is trained or tuned to reduce a specified loss, optimise a reward, or satisfy an externally defined evaluation criterion. The target can be rich and the process can include reinforcement learning, synthetic data, tool use, post-training feedback, and human curation. The phrase does not imply that such systems are literally static or never interact with an environment. It identifies a design tendency: the criteria that count as success are principally fixed outside the adaptive process, and evidence of success is collected against engineered targets.

Reciprocal coupling names a different possible relation. An adaptive system acts in an environment, encounters constraints and novelty, changes its activity, and in doing so changes the local conditions that later constrain it. Enactive and autopoietic traditions treat organism–environment relation as central to cognition (Maturana and Varela 1980; Varela, Thompson, and Rosch 1991; Thompson 2007). This philosophical lineage does not furnish a completed engineering specification, and it should not be treated as a license to anthropomorphise software. It identifies a question: are some forms of resilient, context-sensitive generality more likely where a system can

maintain, inspect, and revise its own relation to evidence, consequences, and environment than where it is only scored against a fixed target?

The reciprocal-coupling hypothesis is therefore conditional. If reliable generality partly depends on continuing reciprocal relation, then systems organised predominantly around bounded objectives and distributions may require additional mechanisms before they demonstrate it. If carefully designed closed optimization reliably meets the same transfer, calibration, adaptation, and recovery tests, the hypothesis is weakened. If host-governed or environment-coupled systems outperform matched alternatives on those tests, it gains support. The proposal is falsifiable only when comparison conditions, task changes, and failure criteria are specified in advance.

The hypothesis should not be mistaken for a claim that feedback alone creates intelligence. Many technical systems receive feedback, update state, and influence their environment without exhibiting the kind of generality at issue. The relevant proposal is more demanding: feedback must be connected to an inspectable organisation of evidence, error, constraint, and revision. A thermostat has a feedback relation; it does not thereby possess open-ended conceptual transfer. A language-model agent can call an external tool; it does not thereby possess a durable account of which claims were verified, which assumptions were used, and which action permissions remain valid. These examples are illustrations of a distinction, not an attempt to set a final boundary around cognition.

There are also alternative explanations of the observed gap. It may arise from insufficient data diversity, benchmark artefacts, token-based representations, optimisation difficulty, inadequate memory, weak tools, or a mismatch between current training regimes and the tested form of transfer. Some of these explanations may be solved by better versions of familiar techniques. Reciprocal coupling is worthwhile only if it yields discriminating predictions beyond the generic advice to add more data, compute, or feedback. The architecture in the next section is offered as one way to formulate those predictions, not as evidence that the hypothesis has already prevailed.

This framing corrects two possible misconceptions. First, it does not classify all present AI as “closed” in an absolute sense. Deployed systems can use tools, memories, human feedback, and external services. The question is whether those components provide durable, inspectable control over evidence and consequences, and whether their contribution is measured. Second, it does not make embodiment a metaphysical prerequisite. Physical embodiment may be valuable for some capacities; the present claim is narrower, concerning a functional relation among action, evidence, constraints, and revision that can in principle be studied in digital environments.

5. A Host-Governed Design Proposal

One way to convert the hypothesis into a researchable proposal is to separate learned possibility generation from durable governance of what a system may treat as evidence or act upon. The proposed architecture is deliberately modular. A learned model supplies candidate descriptions, plans, analogies, and continuations. A host process maintains records, applies explicit constraints, selects context, routes tasks, requests verification, and preserves uncertainty or contradiction where it has not been resolved. This is not a claim that the host is intelligent, that the resulting system is conscious, or that a modular diagram has solved generality. It is an account of separable functions that can be implemented and tested.

The first function is evidence preservation. Exact source material is held separately from summaries, interpretations, and generated hypotheses. A later claim can then be traced back to its source rather than being supported only by a cascade of paraphrases. The second is assumption control. Before an interpretation becomes operational, its premises and scope conditions are recorded, so a system cannot silently rewrite them after observing a convenient answer. The third is bounded comparison. Cases can be linked through explicit relations, but an abstraction is treated as provisional until it survives a defined comparison or test. The fourth is contradiction retention. Unresolved conflicts are recorded as conflicts rather than collapsed into a fluent but unsupported narrative. The fifth is action governance. A candidate action is not treated as authorised merely because a model proposed it; it may require a source check, a rule-based gate, a human approval, or a domain-specific verification process.

These functions are familiar in parts of software engineering, scientific practice, information retrieval, safety engineering, and human oversight. The proposal is not that each is novel in isolation. Its claim is architectural: locating them in an inspectable host layer permits a learned model to remain a fallible generator of possibilities while evidence, authority, and irreversible action are governed elsewhere. That allocation can make failures easier to locate. It can also introduce latency, maintenance costs, false confidence in the host's rules, and new failure modes when records or constraints are incomplete. Those costs belong in any evaluation; a host layer should not be credited with reliability merely because it makes accountability more legible.

The proposal yields empirical questions. Does evidence separation improve factual fidelity or merely slow task completion? Do explicit assumption records reduce post-hoc rationalisation in multi-step work? Does preserving contradiction improve calibrated deferral, or does it overwhelm users with unresolved material? Do approval gates prevent harmful actions at an acceptable cost in time and autonomy? Do explicit comparative structures support transfer to new cases better than a matched model-plus-retrieval baseline? The answers cannot be inferred from the architecture. They require pre-specified tasks, matched comparisons, error analyses, and measures of both benefit and burden.

The relation to present AI systems should be stated carefully. Tool-augmented systems, retrieval-augmented generation, memory systems, workflow engines, and human-in-the-loop applications already instantiate some external functions. The proposal does not deny that. It asks for a stronger separation of authority: generated text or plans are candidates; durable evidence, policy constraints, and action permissions reside in components that can be independently inspected and revised. A system that uses a retrieval tool but allows unsupported output to become an action without a check is not equivalent to a system that makes the check a condition of action. Conversely, an external gate is not automatically meaningful if its rules are irrelevant, stale, or easy to bypass.

This is why the architecture is a research programme rather than a branding exercise. It permits a testable prediction: on tasks involving changing sources, conflicting evidence, and consequential actions, a well-specified host-governed system may produce more traceable and better-calibrated outcomes than a matched learned-model baseline. A negative result would be informative. It might show that the host adds overhead without improving reliability, that the chosen constraints are poorly designed, or that the hypothesised benefit depends on a narrower task class. None of these possibilities is avoided by describing the system as “general.”

An adequate comparison would avoid giving the proposed system an unearned advantage. The baselines should have comparable access to models, tools, source material, compute, and time. The evaluation should distinguish a correct answer produced by luck from an answer supported by a recoverable evidence path. It should include cases where the host constraints are inconvenient or misleading as well as cases where they help. It should test whether a user can override a gate appropriately, because a system that prevents every action is not reliable merely by being inert. And it should report error modes: unsupported assertion, missed evidence, inappropriate deferral, rule conflict, unsafe action, and inability to recover after an incorrect step.

The proposal has a governance consequence even before it has a cognitive one. Separating generation from authority changes what an institution can inspect. A model output can be logged as a proposal; an evidence record can be independently checked; a policy rule can be versioned; an approval can be attributed to a person or process. This does not make the decision correct. It makes the route to decision more contestable. In settings where contestability, appeal, and correction matter, that may be valuable independently of any claim about AGI. In settings where speed is the overriding constraint, it may be impractical. The point is to expose the trade-off rather than to hide it behind a claim that a model “understands.”

6. A Limited Biological Comparison

The design proposal has a restricted functional parallel with a recent biological theory by Miller, Brincat, and Roy (2026). Their theory proposes that stored neural representations may require a faster dynamic control process, mediated by travelling brain waves, to determine which representations become active and how neural populations are coordinated. As the authors’ institutional summary makes clear, the account is theoretical and direct evidence for the proposed analog computation remains an empirical target (MIT Picower Institute 2026).

The comparison does not validate the proposed host architecture. The neuroscience theory concerns biological mechanisms; the architecture proposed here is an explicit digital arrangement. It does not model waves, neurons, consciousness, or biological computation. Nor does the biological theory establish that external software controls are necessary for artificial intelligence. The limited point is functional: both distinguish a stored or generative substrate from a process that selects, routes, and constrains what becomes active. In the biological account, that process is hypothesised to be wave-mediated coordination. In the proposed architecture, it is an auditable host process.

The value of the comparison is therefore heuristic. It motivates a comparative question, not a claim of biological equivalence: can a system with a learned substrate plus independently inspectable controls over evidence, routing, and verification demonstrate more reliable task-dependent behaviour than a matched learned substrate alone? The answer must come from comparative tests, not from analogy.

7. Generality Language and Infrastructural Entrapment

AGI is not a settled technical threshold. Definitions differ in their treatment of breadth, human-relative performance, autonomy, economic value, and physical-world competence. The terminology proposed by Morris et al. (2023) is useful precisely because it separates several of these dimensions rather than offering one undifferentiated label.

Public claims should make comparable distinctions. It is possible to say that a system is highly capable in a defined class of tasks without implying that it is robustly general, autonomous, or reliable in every context of use.

The governance concern is not only semantic. This article calls one possible risk mechanism *infrastructural entrapment*: a capable but bounded system becomes difficult to replace because organisations restructure work around its apparent scale, speed, and cost advantages before its reliability boundaries are sufficiently understood. The term does not assert that every sector has already reached that state, that staff necessarily lose all expertise, or that a system cannot be withdrawn. It identifies a plausible path dependence that deserves scrutiny where institutions embed AI into routines, data flows, procurement contracts, and accountability structures.

The mechanism is compatible with established concerns about automation use and misuse. Parasuraman and Riley (1997) distinguish appropriate use from misuse, disuse, and abuse, and argue that the human factors of automation deserve attention alongside its technical benefits. Their framework does not specifically establish “infrastructural entrapment” for current generative AI. It supports the narrower premise that institutional reliance on automation cannot be assessed from technical performance alone. The contemporary relevance is an inference: when organisations integrate a system across multiple dependencies, recovering independent human or procedural capacity may become more expensive, even if a failure later exposes the system’s limits.

This inference generates concrete questions. Are decisions being delegated beyond the conditions in which reliability was demonstrated? Are independent fallback procedures preserved and rehearsed? Can operators identify a model’s evidential and jurisdictional boundary, or only its visible output quality? Are audit trails sufficient to reconstruct why an action was taken? Do procurement incentives reward throughput more readily than calibrated uncertainty? Under what circumstances can an organisation actually revert to an alternative process? These are empirical governance questions. The article does not claim their answers in advance.

NIST’s AI Risk Management Framework offers a compatible orientation. It treats risk management as ongoing and contextual, emphasising validity, reliability, safety, security, resilience, accountability, and transparency (Tabassi 2023). For high-consequence settings, this implies that a vendor’s claim of broad capability should not substitute for locally relevant evidence, monitoring, escalation paths, and meaningful human authority. “Human in the loop” is not sufficient if the human lacks time, information, or authority to contest the system’s recommendation.

The argument also has a distributional dimension. A system may be broadly useful while producing uneven error burdens across users, contexts, languages, or task types. Average performance can conceal the populations who encounter the boundary conditions most often. That observation does not entail that every system is unjust. It is a reason for evaluations to report where performance has been established, where it has not, and who bears the residual risk when a system is deployed on the basis of a generality claim.

Infrastructural entrapment should therefore be investigated as a sequence rather than a slogan. An organisation first observes local utility. It then standardises a workflow around the system, shifts records and interfaces to suit it, reduces alternative capacity, and begins to treat the system’s output as the normal route to action. At each stage the decision may be reasonable in isolation. The risk arises if cumulative dependence makes later correction costly

while the conditions of reliable performance remain unclear. Countervailing factors also matter: regulation, professional norms, unions, procurement rules, interoperable data, retained training, and well-designed exit arrangements can limit dependency. The term identifies a possible pattern, not an irreversible law.

This perspective changes what responsible deployment evidence looks like. Besides a model card or aggregate performance statistic, an organisation may need a map of decision authority, a description of fallback capacity, a process for handling contested outputs, and a schedule for reassessing whether the system still performs within its validated scope. These are not necessarily new documents for every low-risk use. They are proportionate governance questions. The core issue is whether a claimed capacity is being translated into institutional authority without a corresponding account of error, appeal, and recovery.

8. A Research and Governance Programme

The preceding diagnosis does not recommend abandoning ambitious AI research. It recommends making claims and investments proportionate to evidence. Three research priorities follow.

First, evaluations should test changed conditions alongside average task performance. Relevant dimensions include novel recombination, distribution shift, source quality, tool availability, context limits, conflicting evidence, uncertainty calibration, correction, and recovery after error. A single universal benchmark is unlikely to settle the question. The purpose is not to create a hurdle that no system can cross, but to make the target conditions visible and to distinguish success on one condition from a claim of general success.

Second, research should compare system organisations rather than model families alone. Candidate comparisons include model-only inference, retrieval- and tool-augmented systems, continual-learning systems, human-supervised workflows, and host-governed systems that explicitly separate evidence, constraints, routing, and authority. Fair comparison requires matched tasks, resources, and error criteria. It should measure overhead and operator burden as well as accuracy or completion. A system that improves traceability but makes a time-critical process unusable has not achieved an unqualified improvement.

Third, deployments should retain proportional fallbacks. This does not require a blanket prohibition on automation. The stringency of verification and reversibility should increase with the consequence of error, uncertainty of task conditions, and difficulty of recovery. In low-consequence settings, lightweight oversight may be adequate. In domains involving rights, safety, essential services, or irreversible commitments, organisations should be able to identify the grounds of a recommendation, interrupt an automated process, and use an independent route when the system's demonstrated scope is exceeded.

This agenda is consistent with disagreement about scaling. The AAAI 2025 Presidential Panel survey reported that many respondents considered merely scaling current approaches unlikely or very unlikely to yield AGI (AAAI 2025). A survey of expert views is not technical proof, and it does not determine research outcomes. It does establish that methodological uncertainty is live enough that plural investment is reasonable. The constructive

implication is not to select one philosophical theory as final, but to fund decisive comparisons among specified mechanisms and claims.

The agenda also clarifies what would count against this article's concerns. Robust positive evidence would include pre-specified comparative tests in which systems transfer across materially changed tasks, maintain calibrated uncertainty, recover from correction, preserve accountable evidence trails, and do so without hidden reliance on task-specific human repair. Such evidence would warrant stronger generality claims. The article's position is revisable because its recommendation is evidential rather than metaphysical.

Methodological pluralism has an additional advantage: it turns disagreements that are often framed as forecasts into engineering questions. Instead of asking whether scale will or will not "inevitably" produce AGI, researchers can ask which system boundary transfers under which change; which controls improve calibration at what cost; which memory arrangements retain source fidelity; and which human–AI arrangements preserve meaningful review. Results may favour scaling, external governance, continual learning, embodied interaction, hybrid systems, or different combinations across different tasks. A programme that permits such differentiated answers is more informative than a single label applied to heterogeneous systems.

For public institutions, proportionality is particularly important. A government, university, health service, or employer need not resolve the philosophy of intelligence before deciding how to procure AI. It can ask more immediate questions: what exactly has been demonstrated in a comparable setting; who may contest an output; what happens when a source is wrong; what is the fallback; and who bears responsibility for a failure. These questions do not slow every use case equally. They focus effort where a glossy claim of generality would otherwise be used to bypass the ordinary work of specifying responsibility.

9. Conclusion

Current AI systems are powerful, useful, and likely to remain socially consequential regardless of whether they satisfy any future definition of AGI. That fact makes calibrated language more important, not less. Broad performance and economic usefulness do not by themselves demonstrate systematic generalization, open-world robustness, or durable adaptive organisation. Current research documents both significant limitations and promising interventions. The appropriate conclusion is that the relevant properties remain to be specified and tested.

The reciprocal-coupling hypothesis offers one possible account of why a gap may persist under predominantly closed optimization. It is not a proof of impossibility. The host-governed architecture offers a concrete way to separate learned possibility generation from evidence, constraints, and action authority. It is not a demonstrated cognitive system. The biological comparison is heuristic, not validation. Infrastructural entrapment is a risk mechanism to investigate, not a claim that present deployment has already produced an inevitable outcome.

Those distinctions are the paper's central recommendation. Treat present systems as capable but bounded tools unless a stronger capability has been demonstrated under relevant conditions. Reserve AGI claims for specified, independently tested capacities. And direct some research and governance effort toward architectures and

institutions that can make evidence, uncertainty, verification, and authority explicit. That is not a call to lower ambition. It is a call to make ambition answerable to what the evidence actually shows.

Statements and Declarations

Competing interests: The author declares no competing interests.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used generative AI tools to assist with drafting, structural review, literature targeting, and document preparation. The author reviewed and edited the manuscript and takes responsibility for the submitted content.

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